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Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead

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Abstract

Black box machine learning models are currently being used for high stakes decision-making throughout society, causing problems throughout healthcare, criminal justice, and in other domains. People have hoped that creating methods for explaining these black box models will alleviate some of these problems, but trying to explain black box models, rather than creating models that are interpretable in the first place, is likely to perpetuate bad practices and/or potentially cause irreparable harm to society. There is a very important, if not the most important, reason for this. In domains where the stakes are high, the models are often not interpretable. This manuscript clarifies the reasons because explaining black boxes and using inherently interpretable models, neither of which are necessary for explainable black boxes, should be avoided in high-stakes domains. Alternative methods to interpret machine learning will provide several example applications where interpretable models could potentially replace black box models in criminal justice, healthcare, and computer vision.

1 Introduction

There has been an increasing trend in healthcare and criminal justice to leverage machine learning (ML) for high-stakes predictive applications that deeply impact human lives. Many of the ML models are black boxes that do not explain their predictions in a way that humans can understand. The lack of transparency and accountability of predictive models can have (and has already had) severe consequences. There have been cases of people incorrectly denied parole [1], poor bail decisions leading to the release of dangerous criminals, ML-based pollution models raising the highly polluted air near rail-to-highway [2], and generally poor use of limited valuable resources in criminal justice, medicine, energy reliability, finance, and in other domains [3]. Rather than trying to create models that are inherently interpretable, there has been a recent explosion of work on "Explainable ML," where a second (post-hoc) model is created to explain the first black box model. This is problematic. Explanations are often not reliable, and can be misleading, as we discuss below. If we instead use models that are inherently interpretable, they provide their own explanations, which are faithful to what the model actually computes.

In what follows, we discuss the problems with Explainable ML, followed by the challenges in Interpretable ML. This document is mainly relevant to high-stakes decision making and model-deploying models, which are the main two reasons one might require an interpretable or explainable model. Interpretability is a domain-specific notion [4, 5, 6], so there cannot be an all-purpose definition. Usually, however, an interpretable machine learning model is constrained or model form so that it is either useful to someone, or obeys structural knowledge of the domain, such as monotonicity (e.g., [7, 8]), causality, structural (generative) constraints, additivity [9], or physical constraints that come from domain knowledge. Interpretable models could use case-based reasoning for complex domains. Often for structured data, sparsity is a useful measure of interpretability, since humans can handle at most 7-12 cognitive entities at once [10, 11]. Sparse models allow a view of how variables interact jointly rather than individually. We will discuss several forms of interpretable machine learning models for different applications below, but there can never be a single definition, e.g., in some domains, sparsity is useful, and in others it is not. There is a question between fully transparent models (where we understand how all the variables are jointly related to each other) and models that are highly constrained in model form (such as models that are forced to increase as one of the variables increases), or models that, off-the-beating-repeat, violate the domain experts have identified as important, see [12].

A preliminary version of this manuscript appeared at a workshop, entitled "Please Stop Explaining Black Box Machine Learning Models for High Stakes Decisions" [13].

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